

SAVITRIBAI PHULE PUNE UNIVERSITY

A COMMUNITY ENGAGEMENT PROJECT REPORT ON

CLIMATE CHANGE

**SUBMITTED TOWARDS THE
PARTIAL FULFILLMENT OF THE REQUIREMENTS OF**

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Technology) BY

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CERTIFICATE

This is to certify that the community engagement Project Entitled

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Abstract:

Climate change is one of the most critical global challenges of the 21st century, caused largely by human activities such as industrialization, deforestation, and the burning of fossil fuels. It results in rising global temperatures, melting glaciers, sea-level rise, and frequent extreme weather events. These changes threaten ecosystems, biodiversity, agriculture, water resources, and human health. The project explores the causes, impacts, and possible solutions to climate change, emphasizing the importance of renewable energy, sustainable practices, and international cooperation. By raising awareness and promoting action, this project aims to highlight the urgent need to address climate change and ensure a sustainable future for coming generations.

Climate change represents a serious and accelerating threat to the Earth's environment and human well-being. It is primarily driven by human-induced factors such as the excessive emission of greenhouse gases, deforestation, rapid industrialization, and unsustainable consumption of natural resources. These activities have intensified the greenhouse effect, resulting in rising global temperatures, erratic rainfall patterns, prolonged droughts, stronger cyclones, and unprecedented melting of glaciers and polar ice caps.

The impacts of climate change are far-reaching, affecting agriculture, food security, freshwater availability, biodiversity, and human health. Vulnerable communities, especially in developing nations, are disproportionately affected, facing risks like displacement due to sea-level rise and loss of livelihoods. Climate change also threatens economic stability and global peace by increasing competition for scarce resources.

This project seeks to analyze the causes, consequences, and solutions related to climate change. It emphasizes the urgent need for mitigation strategies such as reducing carbon footprints, adopting renewable energy sources, conserving forests, and promoting sustainable development. Adaptation measures, including disaster preparedness, resilient infrastructure, and international cooperation through frameworks like the Paris Agreement, are equally important

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1. Introduction :

Climate change refers to the long-term alteration of the Earth's climate system, primarily caused by natural processes and, more significantly in recent times, by human activities. It involves shifts in weather patterns, rising global temperatures, melting glaciers, extreme weather events, and changes in rainfall distribution. While the Earth's climate has always changed due to volcanic eruptions, solar radiation, and natural cycles, the rapid changes observed in the past century are largely driven by human-induced factors such as industrialization, deforestation, and the excessive burning of fossil fuels.

The accumulation of greenhouse gases (like carbon dioxide, methane, and nitrous oxide) in the atmosphere traps heat, leading to what is commonly known as the greenhouse effect. This effect, though natural and necessary for life, has intensified due to human activity, resulting in global warming and its widespread impacts on ecosystems, biodiversity, and human societies.



Fig: Climate Change

2. Problem Statement

Climate change represents one of the most critical and urgent challenges facing the world today. Driven primarily by human activities such as the burning of fossil fuels, deforestation, and industrial processes, the Earth's climate is warming at an unprecedented rate. This has led to rising global temperatures, melting polar ice caps, rising sea levels, more frequent and severe weather events (such as hurricanes, droughts, and floods), and disruptions to ecosystems and biodiversity.

Despite growing awareness, there remains a significant gap between scientific understanding and effective policy implementation. Many communities—particularly those in developing countries—are disproportionately affected due to limited resources for adaptation and mitigation. Additionally, there is insufficient education, public engagement, and political will to implement sustainable practices on a large scale.

This project aims to investigate the causes, effects, and potential solutions to climate change, with a focus on increasing awareness, promoting sustainable behaviour, and encouraging policy change through data-driven research and community engagement.

3. Causes of Climate Change :

Mainly it contains two types of causes:

A. Man-Made Causes of Climate Change:

Climate change is primarily driven by human activities that release greenhouse gases (GHGs) into the atmosphere. These gases trap heat and disrupt Earth's natural climate balance. Below are the main human-induced causes:

1. Burning of Fossil Fuels:

The combustion of coal, oil, and natural gas for:

- Electricity generation
- Transportation (cars, trucks, ships, airplanes)
- Industrial processes
- Heating and cooking

❖ How it contributes to climate change:

- Releases carbon dioxide (CO₂) – the most abundant anthropogenic greenhouse gas.
- Releases methane (CH₄) and nitrous oxide (N₂O) in extraction and transport.
- Causes an enhanced greenhouse effect, trapping heat in the Earth's atmosphere.

❖ Examples:

- Power plants (especially coal-fired)
- Gasoline and diesel vehicles
- Industrial boilers and furnaces
- Air travel

2. Deforestation and Land Use Change

The clearing of forests for:

- Agriculture (e.g., cattle ranching, soy, palm oil)
- Urban development and infrastructure
- Logging for timber and fuel

❖ How it contributes to climate change:

- Forests act as carbon sinks – they absorb CO₂ from the atmosphere.
- Removing trees reduces CO₂ absorption and releases the CO₂ stored in trees.
- Affects regional rainfall and increases risk of drought and desertification.

❖ Examples:

- Amazon rainforest deforestation
- Slash-and-burn farming
- Urban sprawl into forested areas

3. Agricultural Practices

Farming activities such as:

- Livestock farming (cattle, sheep)
- Crop cultivation (especially rice paddies)
- Use of synthetic fertilizers and manure

❖ **How it contributes to climate change:**

- **Methane (CH₄) from:**
 - Livestock digestion (enteric fermentation)
 - Manure decomposition
 - Rice paddies (due to anaerobic conditions)
- **Nitrous oxide (N₂O) from:**
 - Fertilizer use
 - Tilling of soil
- **Both gases are much more potent than CO₂:**
 - Methane = ~25x stronger than CO₂
 - Nitrous oxide = ~298x stronger than CO₂

❖ **Examples:**

- Industrial livestock farms
- Overuse of nitrogen fertilizers
- Traditional rice farming

4. Industrial Activities and Manufacturing

Emission of GHGs during the production of goods and materials, including:

- Cement
- Steel
- Chemicals
- Electronics

❖ **How it contributes to climate change:**

- CO₂ released during processes like cement production (from heating limestone).
- Fluorinated gases (e.g., HFCs, PFCs, SF₆) used in refrigeration and manufacturing:
- These are synthetic gases with high global warming potentials (GWPs).
- Aerosols and particulates also affect cloud formation and atmospheric reflectivity.

❖ **Examples:**

- Cement and concrete factories
- Chemical plants
- Refrigerant use in AC and refrigeration systems

5. Waste Management and Landfills

How human waste is handled, including:

- Solid waste in landfills
- Sewage and wastewater treatment
- Open dumping and burning of waste

❖ **How it contributes to climate change:**

- Methane (CH_4) is released during anaerobic decomposition of organic waste in landfills.
- N_2O can be emitted from sewage treatment.
- Burning of waste, especially plastics, releases CO_2 and other harmful pollutants.

❖ **Examples:**

- Municipal landfills
- Poorly managed wastewater treatment plants
- Open waste burning in developing countries

B. Natural Causes of Climate Change :

1. Volcanic Eruptions:

Large volcanic eruptions release gases, ash, and aerosols (tiny particles) into the atmosphere.

❖ Climate Effects:

- Short-term cooling: Sulphur dioxide (SO₂) forms sulphate aerosols in the upper atmosphere, which reflect sunlight and reduce the amount of solar radiation reaching Earth's surface.
- Greenhouse gases: Volcanic eruptions also release carbon dioxide (CO₂), but usually in much smaller amounts compared to human activities.

❖ Example:

- Mount Pinatubo (1991): Caused a global temperature drop of about 0.5°C for over a year.
- Tambora (1815): Triggered the “Year Without a Summer” in 1816, leading to crop failures and famines.

2. Solar Radiation Changes (Solar Variability):

The amount of energy the Earth receives from the Sun is not constant. The Sun goes through cycles of high and low activity (typically 11-year sunspot cycles).

❖ Climate Effects:

- Increased solar activity can cause minor warming.
- Decreased activity can cause slight cooling.
- However, the overall influence of solar variability is small compared to greenhouse gases from human activity.

❖ Example:

- Maunder Minimum (1645–1715): A prolonged period of low solar activity associated with part of the Little Ice Age in Europe.

3.Orbital Changes (Milankovitch Cycles):

Variations in Earth's orbit and tilt over long timescales (thousands to hundreds of thousands of years) alter how much sunlight different parts of the Earth receive.

❖ Climate Effects:

These changes affect the intensity and distribution of sunlight, leading to natural climate cycles such as ice ages and interglacial (warm) periods.

❖ **The Three Cycles:**

1. Eccentricity: Changes in the shape of Earth's orbit (from circular to elliptical), occurring every ~100,000 years.
2. Obliquity: Changes in the tilt of Earth's axis (between ~22.1° and 24.5°), every ~41,000 years.
3. Precession: The wobble in Earth's rotation, occurring every ~26,000 years.

Example:

- The last Ice Age ended about 11,700 years ago, influenced by these cycles.

4. Ocean Currents and Circulation Patterns:

Oceans distribute heat around the planet through currents driven by temperature and salinity differences (thermohaline circulation), as well as wind patterns.

❖ **Climate Effects:**

- Shifts in ocean currents can cause regional and global climate changes.
- Sudden or gradual changes can lead to warming or cooling in various regions.

❖ **Examples:**

- El Niño: A warming of surface waters in the Pacific that disrupts weather worldwide (e.g., floods in some areas, droughts in others).
- La Niña: The opposite of El Niño, often leading to cooler global temperatures.

5. Natural Greenhouse Gas Emissions

Some greenhouse gases are released into the atmosphere naturally, without human involvement.

❖ **Climate Effects:**

- **CO₂, CH₄ (methane), and N₂O (nitrous oxide) can be emitted from:**

Wetlands (methane)

Forest fires

Volcanoes

Example:

- Melting Arctic permafrost can release stored methane, creating a positive feedback loop warming causes more gas release, which leads to more warming.

3. Effects of Climate Change :

- **Rising Temperatures** – Global warming is increasing average surface temperatures.
- **Melting Glaciers & Polar Ice** – Leading to rising sea levels.
- **Extreme Weather Events** – More frequent floods, droughts, cyclones, and wildfires.
- **Threat to Biodiversity** – Loss of species and habitats.
- **Impact on Agriculture** – Reduced crop yields, soil degradation, and food insecurity.
- **Human Health Risks** – Heat strokes, spread of diseases, water scarcity.
- **Economic Consequences** – Damage to infrastructure, increased disaster management costs.



Fig: Effects of Climate Change

4. Solutions to Climate Change :

1. **Renewable Energy** – Use of solar, wind, hydro, and geothermal energy.
2. **Afforestation & Reforestation** – Planting more trees to absorb CO₂.
3. **Energy Efficiency** – Promoting energy-saving technologies and green buildings.
4. **Sustainable Agriculture** – Organic farming, reducing chemical fertilizers.
5. **Waste Reduction & Recycling** – Managing waste to cut methane emissions.
6. **International Cooperation** – Agreements like the Paris Agreement (2015).
7. **Awareness & Education** – Encouraging eco-friendly lifestyles.

5. Global Agreements on Climate Change

1. **Kyoto Protocol (1997)** – The first international treaty that committed developed nations to legally binding targets for reducing greenhouse gas emissions.
2. **Paris Agreement (2015)** – Nearly 200 countries agreed to limit global warming to well below 2°C, with efforts to restrict it to 1.5°C above pre-industrial levels.
3. **Montreal Protocol (1987)** – Focused on phasing out ozone-depleting substances, indirectly helping reduce greenhouse gases.
4. **Glasgow Climate Pact (COP26, 2021)** – Countries pledged to cut methane emissions, phase down coal, and increase climate finance for developing nations.
5. **UNFCCC (1992)** – The United Nations Framework Convention on Climate Change laid the foundation for international cooperation against climate change.
6. **Copenhagen Accord (2009)** – Recognized the need to keep global temperature rise below 2°C and introduced climate finance pledges.
7. **Doha Amendment (2012)** – Extended Kyoto Protocol commitments until 2020, with updated emission reduction targets.
8. **G20 & G7 Summits** – Major economies often set collective goals to cut emissions, accelerate renewable energy adoption, and fund green transitions.
9. **Green Climate Fund (2010)** – Established to support developing nations with financial resources to adapt and mitigate climate change impacts.
10. **Carbon Neutrality Pledges** – Over 130 countries have pledged to reach net zero emissions between 2050–2070 (e.g., EU by 2050, China by 2060, India by 2070).
11. **REDD+ Mechanism** – A UN-backed initiative that provides financial incentives for reducing emissions from deforestation and forest degradation.
12. **Sustainable Development Goals (SDG 13)** – Climate action is one of the 17 SDGs, urging countries to integrate climate strategies into national policies.
13. **Loss and Damage Fund (COP27, 2022)** – Established to provide financial support for vulnerable countries facing severe climate disasters.
14. **Global Methane Pledge (2021)** – Over 100 countries agreed to reduce methane emissions by 30% by 2030.
15. **International Solar Alliance (2015, launched by India & France)** – Promotes solar energy adoption worldwide, especially in developing countries.

6. Impact of Climate Change on India

1. **Agriculture Instability** – Unpredictable monsoon patterns and rising temperatures reduce crop yields, especially rice and wheat, threatening food security.
2. **Water Scarcity** – Melting Himalayan glaciers, irregular rainfall, and groundwater depletion worsen water shortages in several regions.
3. **Heatwaves** – India is experiencing longer and more intense heatwaves, causing health risks, deaths, and reduced labor productivity.
4. **Floods and Cyclones** – Rising sea levels and warming oceans increase the frequency and intensity of floods, cyclones, and coastal erosion in states like Odisha, West Bengal, and Maharashtra.
5. **Biodiversity Loss** – Habitats in the Sundarbans, Western Ghats, and Himalayan regions are under threat, leading to the extinction risk of species like Bengal tigers and snow leopards.
6. **Air Quality Deterioration** – Climate change worsens smog and pollution levels in urban areas, affecting respiratory health.
7. **Impact on Fisheries** – Warmer sea temperatures disrupt marine ecosystems, affecting fish breeding and threatening livelihoods of coastal fishing communities.
8. **Urban Challenges** – Rapid urbanization combined with rising temperatures increases the “urban heat island” effect, making cities hotter and more polluted.
9. **Energy Demand** – Hotter summers increase demand for air conditioning, leading to higher electricity consumption and strain on power supply.
10. **Rural to Urban Migration** – Droughts, floods, and crop failures push rural populations to migrate to cities, causing socio-economic pressures.
11. **Health Risks** – Rising temperatures spread vector-borne diseases like malaria, dengue, and chikungunya to new areas.
12. **Economic Losses** – Climate-induced disasters cause billions of dollars in damage to agriculture, infrastructure, and livelihoods every year in India.
13. **Glacier Retreat in Himalayas** – Faster melting of glaciers threatens long-term water supply for rivers like Ganga, Brahmaputra, and Indus, which millions depend on.

7. Role of Technology in Fighting Climate Change

1. **Renewable Energy Innovations** – Solar, wind, hydro, and geothermal technologies are replacing fossil fuels, providing clean and sustainable alternatives.
2. **Electric Vehicles (EVs)** – Adoption of EVs reduces dependency on petrol and diesel, cutting down harmful emissions from the transport sector.
3. **Carbon Capture and Storage (CCS)** – Advanced technology can capture carbon dioxide emissions from industries and store them underground to prevent release into the atmosphere.
4. **Smart Grids and Energy Efficiency** – Digital smart grids help manage electricity demand, reduce wastage, and promote efficient use of renewable energy.
5. **AI and Big Data for Climate Prediction** – Artificial Intelligence and big data analytics improve climate modeling, early warning systems, and disaster preparedness.
6. **Green Buildings and Smart Cities** – Eco-friendly architecture, energy-efficient appliances, and sustainable urban planning reduce the carbon footprint.
7. **Precision Agriculture** – Drones, IoT sensors, and satellite monitoring help farmers use water, fertilizers, and land more efficiently, lowering emissions from agriculture.
8. **Waste-to-Energy Technology** – Converting waste into biofuels and renewable energy reduces landfill pollution and supports a circular economy.
9. **Geoengineering Research** – Technologies like solar radiation management (reflecting sunlight back into space) and ocean fertilization are being explored to limit global warming.
10. **Blockchain for Sustainability** – Blockchain ensures transparent carbon credit trading, supply chain accountability, and promotes eco-friendly practices.
11. **Water Conservation Technology** – Smart irrigation, desalination plants, and wastewater recycling help mitigate water scarcity caused by climate change.
12. **Public Awareness through Digital Media** – Online platforms, VR simulations, and mobile apps spread awareness and encourage eco-friendly lifestyle changes.

8. Role of Youth & Community

1. **Green Innovation by Students** – Youth can invent eco-friendly technologies like solar gadgets, biodegradable plastics, and energy-efficient tools.
2. **Climate Awareness Campaigns** – Conducting rallies, eco-marathons, and workshops to spread awareness in schools, colleges, and villages.
3. **Social Media Activism** – Using Instagram, YouTube, and reels to promote climate-friendly habits (#ClimateAction, #FridaysForFuture).
4. **Eco-Friendly Startups** – Young entrepreneurs starting businesses in renewable energy, organic products, and sustainable fashion.
5. **Tree Plantation Drives** – Organizing mass afforestation and reforestation programs.
6. **Community Clean-Up Programs** – Local groups cleaning rivers, lakes, beaches, and neighborhoods.
7. **Green School & Colleges** – Youth leading initiatives like banning plastics, rainwater harvesting, and waste segregation on campuses.
8. **Peer Education** – Students teaching fellow students and families about eco-living.
9. **Participating in Policy Making** – Youth representing communities in climate summits, NGOs, and government forums.
10. **Eco-Volunteering** – Volunteering in NGOs, wildlife conservation, and disaster relief activiforum
11. **Sustainable Lifestyle Movements** – Communities promoting cycling, carpooling, cloth bags, and renewable energy adoption.
12. **Innovation in Agriculture** – Youth farmers adopting organic farming, vertical farming, and climate-smart agriculture.

9. Future of Climate Change (If Not Controlled)

1. **Extreme Global Warming** – Average temperatures may rise by 4–6°C by 2100, making many regions uninhabitable.
2. **Permanent Sea-Level Rise** – Coastal cities like Mumbai, Kolkata, New York, and Tokyo may sink under water.
3. **Superstorms & Mega-Cyclones** – More destructive hurricanes and cyclones due to warmer oceans.
4. **Global Food Crisis** – Crop failures, soil infertility, and reduced yields leading to famine.
5. **Water Wars** – Countries fighting over freshwater resources due to scarcity.
6. **Mass Extinction of Species** – 1 in 6 species could go extinct due to habitat destruction.
7. **Tropical Diseases Spread** – Diseases like malaria, dengue, and Zika spreading to new regions.
8. **Glacier Disappearance** – Complete melting of Arctic ice and Himalayan glaciers, affecting rivers like Ganga, Brahmaputra, and Indus.
9. **Climate Refugees** – Millions displaced from low-lying islands and coastal areas.
10. **Economic Collapse** – Billions lost due to disasters, infrastructure damage, and reduced productivity.
11. **Loss of Cultural Heritage** – Ancient sites like Venice, Sundarbans, and Maldives threatened.
12. **Unbreathable Air** – Cities facing extreme air pollution and toxic smog events.

10. Future Vision: Green Planet

1. **100% Renewable Energy** – Global electricity powered entirely by solar, wind, hydro, tidal, and geothermal sources.
2. **Zero-Waste Lifestyle** – Recycling and reusing become a norm, landfills vanish.
3. **Carbon-Neutral Transportation** – Electric vehicles, bullet trains, solar-powered aircraft, and hydrogen-fuel ships.
4. **Smart Eco-Cities** – Cities with vertical gardens, rooftop farming, eco-buildings, and underground waste systems.
5. **Green AI & Technology** – AI systems managing energy use, predicting climate risks, and optimizing resources.
6. **Ocean Clean-Up Revolution** – Large-scale removal of plastics and toxins from oceans to restore marine life.
7. **Rewilding Earth** – Restoring forests, wetlands, and grasslands to bring back extinct species habitats.
8. **Personal Carbon Tracking** – Every citizen tracks their carbon footprint via apps and reduces it.
9. **Space-Based Solar Power** – Satellites transmitting clean solar energy to Earth.
Green Economy – Jobs created in renewable energy, organic farming, and eco-innovation industries.
9. **Sustainable Fashion & Food** – Clothing made from biodegradable fibers; plant-based and lab-grown foods mainstream.
10. **Global Eco-Education** – Environmental studies compulsory in schools, creating eco-conscious generations.
11. **Universal Clean Water Access** – Technology like desalination and rainwater harvesting ensures no water scarcity.
13. **Climate Justice** – Equal support for developing countries to adapt and grow sustainably.
14. **International Green Law** – A global eco-constitution ensuring protection of forests, , and wildlife.

11. Case Studies

1. Himalayan Glacial Melting (India & Nepal)

- Rapid retreat of Himalayan glaciers due to rising temperatures.
- Threatens water supply for millions dependent on rivers like the Ganga, Yamuna, and Brahmaputra.
- Increased risks of glacial lake outburst floods (GLOFs).

2. Cyclone Amphan (2020, India & Bangladesh)

- One of the strongest cyclones in the Bay of Bengal in decades.
- Caused billions in damages, loss of livelihoods, and mass displacement.
- Highlighted the vulnerability of coastal communities to climate-induced disasters.

3. Australia Bushfires (2019–2020)

- Extreme heat and prolonged drought worsened wildfires.
- Destroyed millions of hectares, killed or displaced nearly 3 billion animals.
- Strong evidence linking climate change to increased wildfire intensity.

4. Heatwave in Europe (2019)

- Record-breaking temperatures crossed 45°C in France and Spain.
- Thousands of heat-related deaths.
- Agriculture and energy sectors severely impacted.

12. Conclusion

Climate change is a pressing challenge that threatens the stability of ecosystems, societies, and economies worldwide. It is primarily caused by human activities, but with collective global efforts, its effects can be mitigated. Adoption of renewable energy, afforestation, sustainable development, and strong international cooperation are essential for reducing its impact. The responsibility lies with governments, organizations, and individuals alike. Immediate action is crucial to secure a safe and sustainable future for upcoming generations.

The growing threat of climate change challenges humanity to rethink its relationship with the environment. From melting glaciers to drought-stricken regions, the planet is signaling the urgency of change. Transitioning to sustainable practices, enforcing climate-friendly policies, and raising environmental awareness can slow this crisis. It is a shared duty of all people to act responsibly and protect the Earth. Our actions today will define the world we leave behind tomorrow.

13. References

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